ProteinDance: An AI-Integrated, Structure-Based Pipeline for End-to-End Drug Discovery

Boran Cui[1], Zhenxiong Wen[1]
[1]*Student, The Affiliated High School of Peking University*
Lingfang Tang[2]
[2]*Supervisor, The Affiliated High School of Peking University*

Abstract: Human disease remains a shared global challenge, and the high cost and low translational efficiency of conventional drug development have hindered the realization of precision medicine. With the rise of AI for Science and advances such as AlphaFold 3, discovery costs are being substantially reduced, making precision therapeutics increasingly attainable. Building on this momentum, we introduce ProteinDance, an AI-based, automated drug-design pipeline that integrates target identification, structure acquisition, pocket prediction, virtual screening (ligand retrieval), and Transformer-based de novo compound design with toxicity optimization (ProteinSkier), thereby enabling a low-cost workflow from disease to drug. Recent data indicate that, in 2024, the screening prevalence of depression among Chinese adolescents reached 39.93%, underscoring its broad impact. Against this backdrop, we selected the serotonin transporter (SERT), a protein implicated in depression, as an exemplar target and used ProteinDance to design a new drug candidate for the treatment of depression, which demonstrated favorable performance in in silico toxicity and related evaluations. We contend that ProteinDance offers an AI-driven pharmaceutical pipeline for future research and a reference framework for the field.

KEYWORDS: Structure-based drug discovery; AI-driven molecular design; ADMET prediction; Serotonin transporter allosteric modulation; ProteinDance

Contents

1 Introduction **3**

2 ProteinDance Structure-Based Drug Discovery Pipeline **4**
2.1 Target Identification and AI-Guided Selection 4
2.2 Protein Structure Retrieval and Preparation 5
2.3 Binding-Site Identification by Pocket Prediction 6
2.4 Ligand Retrieval and AI-Assisted Design 7
2.5 Model Architecture for Lead Generation and Optimization - ProteinSkier 8
2.5.1 Reinforcement Fine-Tuning for Novel Molecule Generation and Lead Optimization. 9
2.6 Result Visualization and Analysis 11

3 AI-Driven Discovery and Optimization of an Allosteric SERT Modulator **12**
3.1 Pocket Identification and Key Residues (SERT Allosteric Site S2) . . . 13
3.2 Virtual Screening and Hit Identification 15
3.3 Hit Validation and Fragment Expansion 16
3.4 ADMET Prediction and Drug-likeness Evaluation 18

4 Discussion **20**
4.1 Accelerating Drug Discovery with AI4Science 20
4.2 Toward Low-Cost Precision Medicine 20
4.3 Limitations 21
4.4 Future Outlook and Applications 21

1 Introduction

Despite substantial scientific advances, developing a single new medicine still requires hundreds of millions of dollars and more than a decade of work, with attrition rates above 90 % between pre-clinical discovery and regulatory approval. [1] This inefficiency stalls the promise of precision medicine—matching the right treatment to the right patient—and leaves many indications underserved. Consequently, lowering discovery costs and raising translational efficiency has become a strategic priority across academia and industry.

The emerging discipline of AI for Science (AI4S) seeks to tackle these bottlenecks by embedding machine-learning models into every stage of the scientific workflow.[2] Landmark breakthroughs such as AlphaFold 3 [3], which predicts multimolecular complexes with up to two-fold higher accuracy than earlier methods, exemplify the field's transformative potential. Parallel advances in chemical language models and transformers now enable property-conditioned molecule generation, virtual screening at proteome scale, and rapid hit-to-lead optimization. The COVID-19 pandemic offered a real-world stress test: AI-driven "virtual labs" iteratively designed stabilized antigen constructs, while pharma–tech alliances automated mRNA sequence engineering and data analysis.[4] Algorithms such as LinearDesign further demonstrated that deep-learning-guided codon rearrangement can raise mRNA stability in minutes, accelerating vaccine prototyping.[5] These case studies provide compelling evidence of the transformative role of artificial intelligence in pharmaceutical research and development, highlighting its substantial contributions and promising future implications.

However, despite the fact that individual pharmaceutical modules in AI4S have already been well developed, to date no effective, all-in-one AI-driven drug discovery pipeline has been proposed, and an integrated end-to-end solution remains lacking. To exploit these opportunities in AI4S, we created ProteinDance, a modular AI-agent pipeline that automates (i) target annotation and structure retrieval, (ii) pocket ranking and physics-based prescreening, (iii) transformer-driven de-novo generation with novelty and drug-likeness constraints, and (iv) multitask deep-learning toxicity filtering. State-of-the-art predictive ADMET toxicology frameworks—integrating omics data and large-scale assay read-outs—provide early flags for idiosyncratic liabilities and reduce downstream animal testing. [6]

Besides, mental-health disorders are poised to become the leading global cause of non-fatal disease burden, and recent surveillance reveals a 39.93 % prevalence of depressive symptoms among Chinese primary and middle-school students. [7] Central to MDD pathology is dysregulated serotonergic signaling; reduced serotonin re-uptake via the membrane serotonin transporter (SERT) correlates with mood disturbance, and SERT remains the pharmacological locus of first-line selective serotonin-reuptake inhibitors (SSRIs). [8] High-resolution SERT structures and in silico docking campaigns have opened new avenues for allosteric or subtype-selective ligands that could deliver faster onset and fewer side-effects than traditional SSRIs. Based on this premise, we further applied the developed ProteinDance to the design of SSRIs drugs that inhibit the SERT protein.[9]

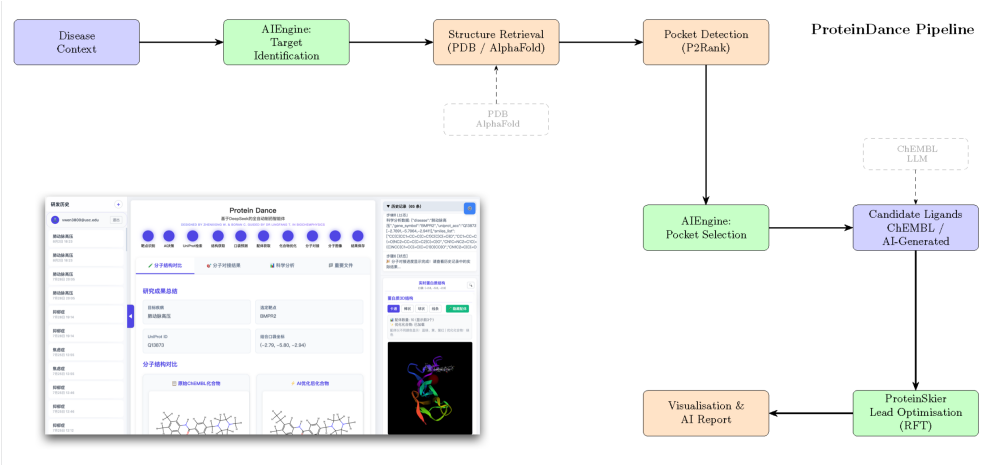

Fig. 1 ProteinDance Pipeline Structure

2 ProteinDance Structure-Based Drug Discovery Pipeline

ProteinDance implements an end-to-end, structure-driven drug discovery workflow that integrates modern AI reasoning with classical bioinformatics tools. The pipeline proceeds through sequential modules for target identification, structure modeling, pocket detection, ligand selection, and lead optimization. At each stage, domain-specific algorithms (e.g. protein structure prediction, binding-site analysis) are coupled with AI-driven decision-making. This design allows the system to automatically find and evaluate a protein target, obtain its 3D structure, identify ligandable sites, propose or retrieve candidate molecules, and optimize a lead compound – all in a unified, automated framework. Key innovations include the use of transformer-based AI engines for context-aware decision support and molecule generation, the incorporation of structural biology data (experimental and predicted protein structures) at the core of the pipeline, and the integration of machine learning models (e.g. for pocket prediction) that respect the 3D geometry (invariant under rotations/translations) of protein structures. Below, we detail each stage of the ProteinDance pipeline in a methods-style description and in Figure 1.

2.1 Target Identification and AI-Guided Selection

The pipeline begins with target identification for a given disease input. An AI engine (the AIEngine module) analyzes the disease context to propose a list of candidate protein targets (gene symbols) associated with the pathology. This AI engine is built on a GPT-style large language model and is prompted to return up to N relevant targets, taking into account an "innovation level" parameter that balances well-established versus novel targets. Internally, the model generates potential targets and their novelty scores by leveraging biomedical knowledge. The result is a set of candidate target gene symbols, each potentially ranked by an innovation or novelty score.

Once candidate targets are identified, the pipeline performs AI-guided target selection. The list of options is passed back into the AI engine to choose the most promising target for drug discovery . The AI considers the disease context and the role of each protein, evaluating which target might be most critical or "druggable" for the disease. This decision is made via a prompt-driven analysis where the AI model compares the target options and provides a recommendation with an explanation . The selected target's gene symbol (e.g., TP53, EGFR) is then carried forward. In parallel, the AI can also generate a textual explanation for why this target was chosen, which can later be presented to the user for transparency. By using a transformer-based AI to drive target selection, the pipeline injects expert knowledge and reasoning at the outset, ensuring that the chosen protein is both relevant to the disease and suitable for a structure-based approach.

2.2 Protein Structure Retrieval and Preparation

After target selection, ProteinDance obtains a three-dimensional protein structure for the target, which is essential for structure-based drug design. The pipeline's PharmaEngine module first maps the gene symbol to a UniProt accession ID by querying the UniProt database [10]. This yields the specific protein identifier (and name) corresponding to the target gene in human (or the relevant organism). Given the UniProt ID, the pipeline then searches for any available experimental structures in the Protein Data Bank (PDB). It uses the RCSB API [11] to find PDB entries associated with the protein's UniProt accession . If one or more PDB structures exist, the system will prefer those, as experimental structures can provide high-resolution detail. In fact, if multiple PDB entries are found, the AI engine is invoked to select an optimal structure among them. For example, the AI might be prompted with the list of PDB IDs and contextual information (resolution, presence of co-crystallized ligands, etc.) and asked to choose the structure most suitable for drug design (favoring high resolution or relevant conformations) . This AI-guided choice ensures that the pipeline picks a high-quality structural template when multiple options are available.

If no experimental structure is available (or if the PDB retrieval fails), the pipeline automatically falls back to an AlphaFold-predicted structure. The PharmaEngine contacts the AlphaFold structure database [3] to download the predicted 3D model for the protein of interest. In the ProteinDance implementation, the UniProt accession is used to fetch the corresponding AlphaFold model (for example, retrieving the AF-¡UniProt¿-F1-model_v4.pdb file). This ensures that even "undruggable" or novel targets without PDB entries can be analyzed in structural detail. The retrieved structure file (PDB format) is saved to a working directory for subsequent steps. In summary, this stage yields a protein 3D structure – either an experimental crystal/NMR structure if available, or a high-confidence AlphaFold de novo model – which becomes the basis for downstream pocket and ligand analysis. Importantly, AlphaFold's advanced neural-network-based predictions (which incorporate evolutionary transformers and equivariant geometry modeling) are thus integrated into the pipeline, providing a structural foundation for targets that lack experimental data.

Before moving on, the chosen structure may be processed or validated (ensuring the file exists and is in the correct format). The pipeline encapsulates the structure info

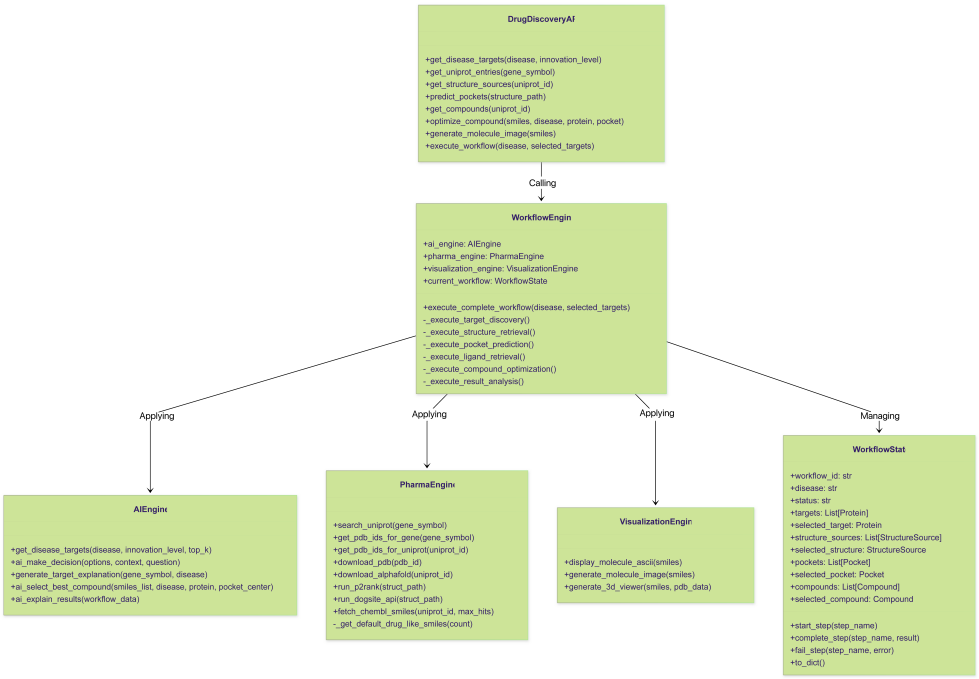

Fig. 2 Function Calling in ProteinDance

for logging and for use in the front-end visualization. By combining database queries and AI decisions, this module automates what would traditionally be a manual search for suitable protein structures. The structure is shown in Figure 2

2.3 Binding-Site Identification by Pocket Prediction

With a protein structure in hand, ProteinDance next identifies potential ligand-binding pockets on the protein's surface. This is accomplished using P2Rank, a machine-learning-based pocket detection tool integrated into the pipeline. [12] P2Rank analyzes the 3D coordinates of the protein to predict "hot spots" where a small molecule could bind. It does so by scoring and clustering points on the protein's solvent-accessible surface, using a model trained on known protein-ligand complexes . In our pipeline, the PharmaEngine calls the P2Rank executable (prank predict) on the target protein's PDB file to generate a ranked list of predicted pockets. The output includes, for each pocket, a geometric center (3D coordinates) and a confidence score reflecting its predicted ligandability. P2Rank's ML model effectively evaluates local surface features to estimate how likely each cavity could bind a ligand, achieving high success rates in identifying true binding sites .

The pipeline parses P2Rank's results (from the output CSV file) to extract the pocket centers and scores. Typically, a handful of top-scoring pockets are considered. At this point, ProteinDance again leverages the AI engine for pocket selection. The

list of candidate pockets (with their scores and coordinates) is fed into the AIEngine's decision function, which is prompted to choose the most suitable binding site for drug development . The AI is given context such as: higher pocket scores indicate a more ligand-friendly site, and an ideal pocket might be one that is druggable (sufficient size, appropriate depth) and relevant to the protein's function or active site. The AI then returns a selected pocket (typically the one with the best characteristics) along with a rationale . By doing this, the pipeline goes beyond automatically trusting the top P2Rank prediction – it uses an AI reasoning layer to ensure the pocket choice aligns with domain knowledge (for example, excluding a high-scoring pocket that is too solvent-exposed or choosing a slightly lower-scoring pocket that is in a known active site region). The coordinates of the chosen pocket's center are recorded for use in subsequent steps (e.g. highlighting the binding site for visualization or focusing ligand placement around that site).

In summary, this stage marries structure-based analysis with AI: a geometric ML model (P2Rank) rapidly scans the 3D structure for pockets, and a context-aware AI refines the selection of the pocket to target. This ensures the pipeline identifies a credible binding site on the protein where ligand intervention is feasible.

2.4 Ligand Retrieval and AI-Assisted Design

Once a binding pocket is defined, the pipeline turns to ligand identification – finding or creating small molecules that could bind in that pocket. ProteinDance first attempts a data-driven approach: querying the ChEMBL database for known ligands of the target protein.[13] The PharmaEngine uses the target's UniProt ID to search ChEMBL (a large repository of bioactive compound data) and fetch a list of reported small-molecule ligands or inhibitors for that protein. Specifically, it retrieves SMILES strings of compounds associated with the target (e.g. any molecule with measured activity against the protein). If successful, this yields a set of candidate known compounds – typically drugs or tool compounds that have affinity for the protein. These compounds serve as starting points (they are likely to fit the binding site since they've been tested on the target or homologs).

However, for novel targets or cases where ChEMBL returns few or no hits, the pipeline employs an AI-driven generative step to ensure a supply of candidate ligands. In this scenario, the AIEngine invokes a ligand design function that uses a generative transformer model to create new chemical structures . The AI is prompted as a medicinal chemist: given the protein target (and disease context), design a number of novel, drug-like molecules that could plausibly bind the target. The LLM model then outputs a list of candidate SMILES strings, representing chemically valid structures that respect drug-likeness guidelines (the prompt explicitly asks for compounds following Lipinski's rule-of-five and with potential activity) and are tailored to the target context. This step effectively performs de novo drug design by leveraging the creativity of a large language model trained or instructed in chemistry. The pipeline validates and filters the AI-generated SMILES to ensure they are chemically valid. If the AI generation also fails or yields no valid molecules (a rare case), the system can resort to a small default library of drug-like compounds as a last fallback .

By combining database mining with AI molecule generation, ProteinDance guarantees a set of candidate ligands for virtually any target. This list may include known reference compounds and novel structures suggested by the AI. Each candidate is represented by its SMILES notation and tracked internally as a Compound object with metadata (source = "chembl" or "AI-generated") . The result of this stage is a portfolio of potential ligands (typically on the order of 10–20 molecules) that will be evaluated in the next step.

2.5 Model Architecture for Lead Generation and Optimization - ProteinSkier

At the core of the AI-guided lead compound selection and optimization stage is a Transformer-based autoregressive generative model operating on SMILES strings, which named ProteinSkier, the architecture shown in Figure 3. The base model uses a GPT-2 style architecture with $N_{\text{layer}} = 12$ transformer blocks, hidden dimension $d_{\text{emb}} = 768$, and $N_{\text{head}} = 12$ attention heads, giving a capacity comparable to GPT-2 Base [14]. Input molecules are represented as character-level SMILES sequences. A custom vocabulary is constructed from the training corpus, encompassing standard SMILES characters (letters, digits, bond/branch symbols such as '()[]+=#%@') as well as special tokens for end-of-sequence/`<|endoftext|>` (serving also implicitly as start), padding `<|pad|>`, and unknown characters `<|unk|>`. The resulting vocabulary size is approximately $|\mathcal{V}| \approx 2000$, dominated by frequent chemical tokens and a small set of control tokens.

Formally, for a SMILES sequence $x = (x_1, x_2, \ldots, x_T)$, the model defines a probability distribution over sequences via the autoregressive factorization:

$$P_\theta(x) = \prod_{t=1}^{T} P_\theta(x_t \mid x_{<t}), \tag{1}$$

where each conditional probability is produced by the Transformer after processing the prefix $x_{<t}$. During pre-training the model is optimized to maximize the likelihood of a large corpus of cleaned molecules (here, canonical SMILES from ChEMBL and supplementary curated datasets) by minimizing the cross-entropy loss:

$$\mathcal{L}_{\text{pretrain}}(\theta) := -\mathbb{E}_{x\sim D} \sum_{t=1}^{T_x} \log P_\theta(x_t \mid x_{<t}), \tag{2}$$

with D the training dataset and T_x the length of sequence x. Sequences are truncated or padded to a fixed maximum length (e.g., 128 tokens) to align with the model's positional capacity. Dropout ($p = 0.1$) is applied on attention and feed-forward sublayers for regularization. After this unconditional pre-training, the prior model learns SMILES syntax and reproduces the general chemical distribution of the training data: sampling from it yields predominantly valid molecules, many resembling known examples, but with limited inherent novelty due to interpolation in learned chemical space.

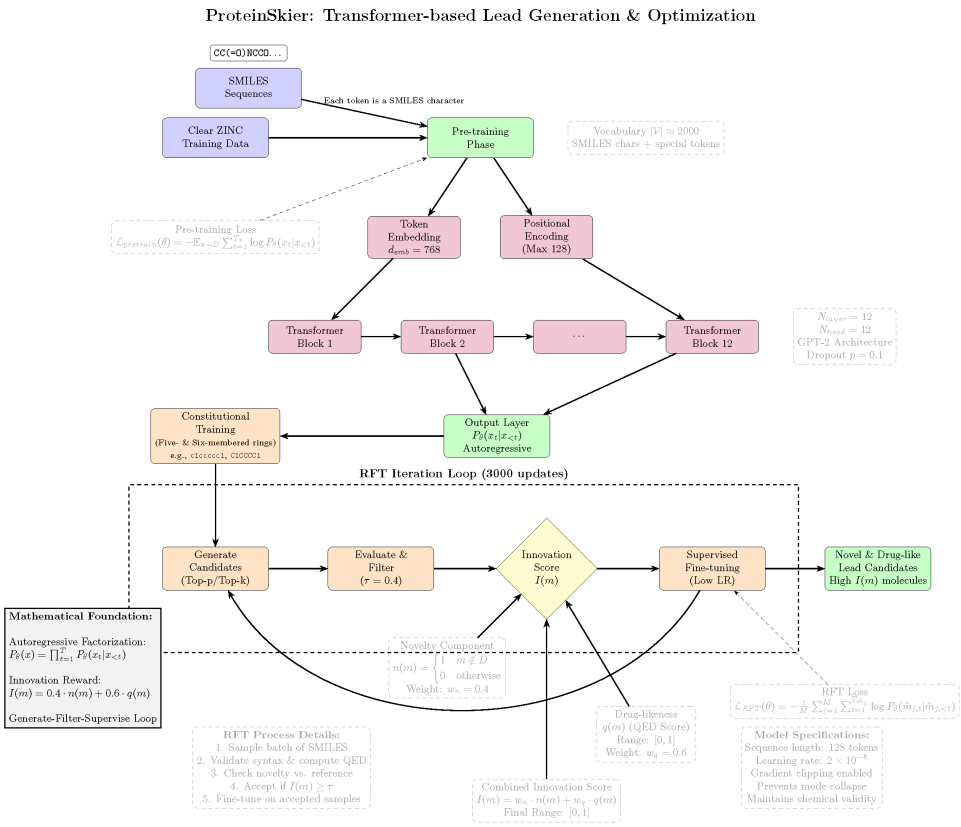

Fig. 3 Protein Skier Architecture Diagram

2.5.1 Reinforcement Fine-Tuning for Novel Molecule Generation and Lead Optimization.

To drive the model toward generating both novel and drug-like lead candidates, we apply a Reinforcement Fine-Tuning (RFT) procedure. Rather than using high-variance policy gradient estimators, we employ a generate-filter-supervise loop: the current model generates candidate molecules, high-quality samples are selected according to a reward function, and the model is then fine-tuned on those examples to shift its generative distribution toward desirable regions of chemical space.

Reward Construction. For each generated molecule m (represented as a SMILES string), we compute two components: a binary novelty indicator $n(m)$ and a continuous drug-likeness score $q(m)$. Novelty is defined as

$$n(m) = \begin{cases} 1 & \text{if } m \notin D \text{ (not present in the reference corpus)}, \\ 0 & \text{otherwise}, \end{cases} \tag{3}$$

ensuring that molecules already seen during pre-training do not receive the novelty bonus. Drug-likeness $q(m)$ is quantified via the QED metric [15], which aggregates desirable physicochemical properties into a score in $[0, 1]$. We combine these into a scalar innovation reward:

$$I(m) := w_n \cdot n(m) + w_q \cdot q(m), \tag{4}$$

with weights $w_n = 0.4$ and $w_q = 0.6$ (empirically chosen to slightly favor drug-likeness over raw novelty), so that $I(m) \in [0, 1]$. A novel molecule with high QED achieves the highest reward; known molecules can only score via their drug-likeness component, thereby discouraging collapse back to memorized scaffolds.

Generate-and-Filter Loop. RFT proceeds for a fixed number of iterations (e.g., 3,000 updates); each iteration consists of:

1. **Candidate Generation:** The current model stochastically samples a batch of SMILES sequences using diversity-preserving decoding strategies (e.g., top-p or top-k sampling), initializing from the begin token and terminating upon end-of-sequence or max length. This yields a set $\{m_i\}_{i=1}^{N}$ of candidates.
2. **Evaluation and Selection:** Each m_i is validated for syntactic correctness (valid SMILES), assigned a QED score, and tested for novelty against the reference set. The innovation score $I(m_i)$ is computed; candidates satisfying $I(m_i) \geq \tau$ are accepted, where τ is a threshold (initially set to 0.4). To avoid stalling when too few candidates pass, τ may be adaptively lowered, but in practice we kept τ fixed to maintain stable pressure toward high innovation.
3. **Supervised Fine-Tuning:** Denote the accepted set as $A_k = \{\tilde{m}_j\}_{j=1}^{M}$. The model is updated by minimizing the cross-entropy loss over these high-reward samples:

$$\mathcal{L}_{\text{RFT}}(\theta) := -\frac{1}{M} \sum_{j=1}^{M} \sum_{t=1}^{T_{\tilde{m}_j}} \log P_\theta(\tilde{m}_{j,t} \mid \tilde{m}_{j,<t}), \tag{5}$$

   with a reduced learning rate (e.g., 2×10^{-5}) and gradient clipping to prevent destabilization. Low-reward or invalid molecules are ignored rather than penalized directly, effectively reweighting the model's distribution toward desirable outputs.
4. **Iteration:** The updated model is used for the next generation batch, and the cycle repeats.

This approach approximates reward maximization by treating top-scoring samples as pseudo-ground truth, enabling stable, example-driven refinement without the complexities of explicit policy gradients or baselines [16]. The explicit novelty term $n(m)$ guards against mode collapse: only unseen molecules contribute the novelty bonus, so the model does not get fine-tuned on memorized structures unless they independently have exceptionally high drug-likeness. Simultaneously, by weighting QED more heavily, the procedure avoids amplifying chemically implausible novel structures, ensuring the generated leads remain in a synthetically and pharmacologically reasonable regime. Secondary filters (e.g., Lipinski violations, synthetic accessibility

heuristics) are monitored post hoc to validate that optimization does not sacrifice downstream developability.

Outcome and Integration. Over successive RFT iterations, the model shifts its generative distribution toward molecules that are simultaneously novel and drug-like. The resulting candidates serve as optimized lead proposals: the system can either directly select high-I molecules as lead compounds or further present them to the downstream decision layer (the AIEngine acting as a medicinal chemist) for final ranking and contextual rationale. The transparency of the process is preserved by retaining the innovation score, the raw SMILES, and the model's internal confidence, allowing human reviewers to inspect why a lead was surfaced. Compared to classic RL-based molecular optimization (e.g., policy gradient frameworks such as REINVENT [17]), this hybrid self-training mechanism stabilizes learning by grounding updates in valid concrete examples while still incentivizing exploration of new chemical space.

2.6 Result Visualization and Analysis

In the final stage, ProteinDance generates outputs for interpretation and reporting. A VisualizationEngine module produces a 2D structural image of the chosen lead compound. This is done by using RDKit [18] to render the SMILES into a molecular diagram. The resulting image (in Base64 or similar format) can be directly embedded in the results. The pipeline also supports 3D visualization: the previously identified binding pocket coordinates and the protein structure can be forwarded to a front-end viewer (e.g. a 3D protein viewer) to illustrate where the ligand would bind. While the pipeline does not perform full docking in this version, it provides the pocket location so that a user could hypothetically place the ligand there for visualization.

Crucially, the pipeline also generates a scientific analysis report via the AIEngine. After completing the workflow, a structured summary of the entire process is compiled – including the disease, selected target and why it's important, the source of the protein structure, details of the binding pocket, and characteristics of the chosen compound. The AI then produces a written narrative that contextualizes these results. For example, it may explain how the target is implicated in the disease, why the particular pocket was chosen (e.g. "it is the active site where endogenous ligands bind"), and why the lead compound is promising (e.g. "it fits well in the pocket and meets drug-likeness criteria"). This explanation is generated by the same AI engine, prompted with the assembled workflow data . The text is intended to read like a brief discussion or rationale section in a research report, thereby closing the loop with human-readable insights.

Finally, the pipeline outputs include: the optimized lead molecule (SMILES and image), the target and pocket information (such as the UniProt ID, PDB/AlphaFold structure used, pocket coordinates), and the AI-generated explanation of the findings. These results are packaged in a JSON response for programmatic use and also displayed in a user-friendly format on the front-end interface. The session can be saved so that all intermediate decisions and results are retained for review.

In summary, Figure 1 shows the structure-based ProteinDance pipeline orchestrates a multi-step drug discovery workflow using a combination of AI and bioinformatics modules. It progresses from disease to target, target to structure, structure

to pocket, pocket to ligand, and ligand to optimized lead, employing advanced methods at each juncture: large language models (transformers) for decision-making and hypothesis generation, protein structure prediction (AlphaFold) for 3D models (leveraging networks with rotational equivariance), pocket detection algorithms (P2Rank) for structural binding-site analysis using machine learning , and medicinal chemistry rules for evaluating compounds. The tight integration of these components within an automated workflow is a key innovation. By preserving human-like reasoning through the AI (for choices and explanations) and grounding those decisions in structural data, ProteinDance exemplifies an AI-integrated, structure-based approach to accelerate early-stage drug discovery. The pipeline's modular design (AIEngine, PharmaEngine, etc.) also ensures it can be extended or refined as new algorithms or models become available, keeping it at the cutting edge of computational drug discovery research.

3 AI-Driven Discovery and Optimization of an Allosteric SERT Modulator

Recent epidemiological data from 2024 report that the screening prevalence of depression among Chinese adolescents reached 39.93%, underscoring the urgent public-health need for more effective and mechanistically diverse antidepressant strategies.[7] To address this, we chose the serotonin transporter (SERT)—a central regulator of synaptic serotonin and a well-established depression-related target—as an exemplar for method demonstration. Leveraging the ProteinDance structure-guided AI-augmented discovery pipeline, we carried out target-informed pocket identification, hit finding, fragment-based lead elaboration, and generative optimization to design a novel allosteric modulator.

Serotonin transporter (SERT) is a transmembrane protein responsible for the reuptake of synaptic serotonin (5-HT) and is the primary target of selective serotonin reuptake inhibitors (SSRIs) used as antidepressants. The human SERT structure has been resolved by X-ray crystallography, revealing two distinct ligand binding sites: the central orthosteric site (S1) where SSRIs like (S)-citalopram bind, and a secondary allosteric site (S2) located in the extracellular vestibule [19]. Binding at the S2 site can modulate the dissociation kinetics of S1 ligands [20], offering an additional mechanism to regulate transporter function. This dual-site mechanism suggests a novel strategy for antidepressant design [21]: ligands targeting the allosteric S2 pocket could synergistically enhance or prolong S1 inhibition, leading to new therapeutic profiles. However, no known drugs directly target the S2 site, making it an attractive yet under-explored target for novel SERT modulators. The mechanism diagram is shown in Figure 4.

Recent advances in artificial intelligence (AI) and machine learning have accelerated drug discovery by enabling efficient exploration of protein pockets and chemical space [22]. In particular, AI tools can assist in identifying cryptic binding pockets, screening large libraries for hits, and even generating novel molecular structures with desired properties. In this study, we leverage the ProteinDance AI-augmented, structure-based workflow to discover and optimize new SERT allosteric modulators. First, we applied geometric pocket prediction and ranking to the SERT structure to identify a candidate allosteric cavity, followed by large-scale virtual screening to

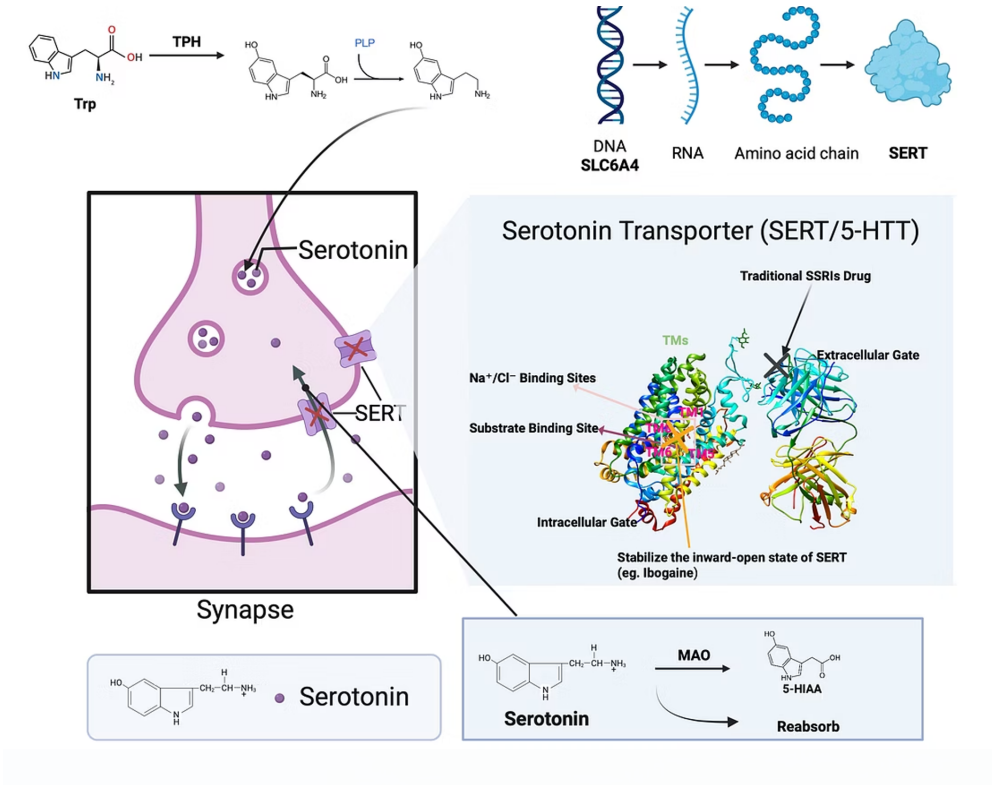

Fig. 4 Mechanism of SERT and SSRIs

obtain initial hit compounds [23]. We then performed fragment-based lead optimization guided by deep learning suggestions to improve target engagement. Building on the refined scaffold, a Transformer-based generative model was employed to propose analogue molecules, with a reinforcement fine-tuning strategy biasing generation toward high novelty and drug-likeness. By tightly integrating structure-based design with generative chemistry and AI-guided decision making, our pipeline aims to yield potent, novel, and pharmacologically credible allosteric SERT modulators.

3.1 Pocket Identification and Key Residues (SERT Allosteric Site S2)

Using ProteinDance's structure-based workflow to orchestrate pocket discovery, we applied P2Rank (via the PrankWeb-style pocket identification module) to the SERT crystal structure (PDB 5I6Z) and identified a prominent ligand-binding cavity in the extracellular vestibule, spatially distinct from the central orthosteric S1 site. ProteinDance parsed and ranked the predicted pockets by calling p2rank API [12], selecting this cavity based on its high ligandability score (confidence score = 23.96, placing it above approximately 87% of random surface cavities). The pocket's geometric center was computed to be near coordinates (45.13, 178.22, 141.59) in the SERT coordinate

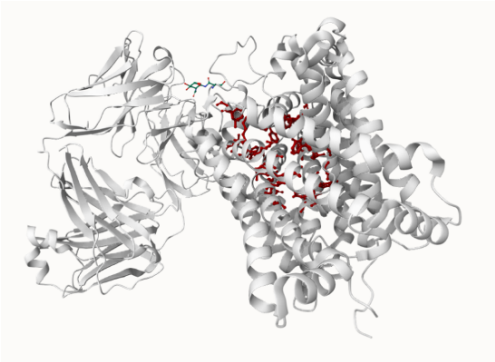

Fig. 5 SERT structure (PDB 5I6Z) with the ProteinDance-identified S2 allosteric pocket highlighted relative to the central S1 binding site.

frame, with an effective radius of 9 Å (roughly covering an 18 Å cubic region) as extracted and propagated by the pipeline. Notably, no SSRI drugs in existing SERT co-crystal structures occupy this region, supporting the interpretation that the site corresponds to the allosteric S2 pocket rather than the canonical S1 site. Figure 5 shows the SERT structure with the ProteinDance-predicted S2 pocket in relation to the central site.

We selected this novel pocket as the focus for downstream virtual screening through ProteinDance, reasoning that an allosteric modulator binding here could provide a mechanism of transporter regulation that is non-competitive with S1-binding SSRIs, potentially yielding distinct pharmacology or reduced side effects in combination therapies.

Structural inspection of the SERT crystal structure (PDB 5I6Z), contextualized by prior mutagenesis and structural literature and integrated within ProteinDance's interaction-mapping module, highlighted a triad of charged residues—Arg104, Asp328, and Glu494—as principal interaction hotspots within the S2 vestibule. Arg104, positioned at the extracellular entrance of the pocket, behaves as a positively charged "gatekeeper"; its salt bridge with Glu494 likely undergoes transient disruption to permit ligand ingress, and its guanidinium moiety presents a cation–π interaction platform for aromatic substituents. Asp328, deeper within the pocket, contributes the negative electrostatic anchor typically engaging protonated amines of ligands, exerting dominant influence on affinity. Glu494, on transmembrane helix 10, complements Arg104 to form the dynamic salt-bridge gate and participates in water-mediated hydrogen bonding that stabilizes incoming ligands. Together, these residues define an electrostatic and conformational landscape that any S2-directed chemotype must negotiate; their coupled motions appear capable of transmitting allosteric signals toward the central binding core.

By centering the subsequent virtual screening on this pocket and explicitly prioritizing interactions with Arg104, Asp328, and Glu494, we aimed to identify compounds that could either perturb the Arg104–Glu494 gate or bridge to Asp328, thereby exerting an allosteric modulation of SERT. These three residues were thus encoded as key

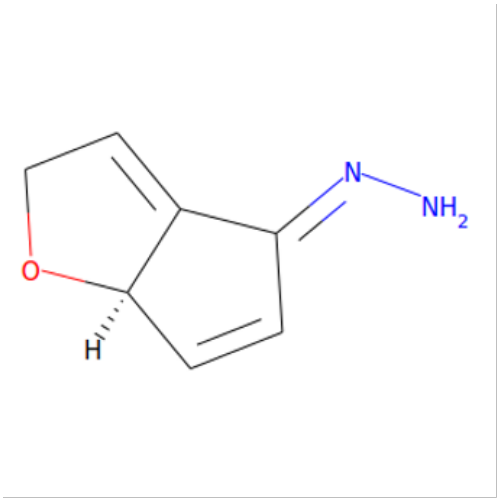

Fig. 6 Chemical structure of *24340347_Intermediate* with core fragment highlighted for further growth.

interaction targets in the docking selection criteria, ensuring predicted hits engaged the pocket in a functionally meaningful manner.

3.2 Virtual Screening and Hit Identification

Leveraging ProteinDance's docking orchestration layer, we conducted a large-scale virtual screen using MTiOpenScreen of the Diverse-lib compound collection ($\approx 1.5 \times 10^5$ drug-like molecules) against the previously defined S2 pocket. The docking grid was automatically centered at the ProteinDance-computed pocket centroid (45.13, 178.22, 141.59), with the volume defined to encompass Arg104, Asp328, and Glu494. AutoDock Vina was employed to score protein–ligand poses; during scoring, soft distance-based biases were introduced to favor ligand placement proximal to the charged residues, encouraging formation of salt bridges and cation–π interactions in accordance with the key residue map.

From this screen, the top-ranked candidate—designated *24340347_Intermediate* (SMILES: `N/N=C1\C=C[C@H]2OCC=C12`), shown in Figure 6 —emerged with a Vina-predicted binding affinity of **-11.6 kcal/mol**, a value considerably more favourable than competing compounds. This score suggests a stable engagement in the S2 pocket (more negative values denote tighter predicted binding), and is notable for an allosteric site, approaching or exceeding affinities typical of orthosteric S1 inhibitors, implying a high degree of ligand–protein complementarity.

Table 1 summarizes physicochemical and docking descriptors of the top hit:

These values satisfy standard "Hit-like" heuristics (MW < 400, HBD ≤ 5, HBA ≤ 10, moderate lipophilicity), indicating a favorable size/polarity balance and synthetic tractability [15]. The intermediate's moderately hydrophobic character (logP ≈ 3.5) enables penetration into the hydrophobic vestibule while retaining sufficient aqueous solubility. Structural interpretation of the docking pose reveals an aromatic

Table 1 Key physicochemical and docking parameters of the selected hit.

Parameter	Value
Binding affinity (AutoDock Vina)	$-11.6\,\mathrm{kcal\,mol^{-1}}$
Molecular weight	$\sim 397\,\mathrm{Da}$
Hydrogen-bond acceptors (HBA)	6
Hydrogen-bond donors (HBD)	1
Rotatable bonds	4
Predicted $\log P$	~ 3.5

core engaging hydrophobic regions, with peripheral polar functionalities (putative amine/amide/hydroxyl moieties) poised to form hydrogen bonds or salt bridges with polar pocket residues, notably Asp328 and Glu494—consistent with the mixed hydrophobic/polar nature of the S2 site.

The exceptional score of *24340347_Intermediate* in the Diverse-lib screen supports the capacity of the SERT S2 allosteric cavity to accommodate a moderately sized, amphipathic aromatic chemotype. Pose analysis indicated: (i) the putative protonated center is positioned within salt-bridge distance of Asp328, echoing its known anchoring role; (ii) an aromatic substituent aligns toward Arg104 in a geometry compatible with cation–π contacts; and (iii) the ligand fills a substantial fraction of vestibular volume while leaving a small adjacent void, suggesting clear vectors for fragment-based elaboration.

3.3 Hit Validation and Fragment Expansion

Analogue Search via SwissSimilarity:

As part of ProteinDance's analogue and scaffold exploration subroutine, we queried SwissSimilarity with the SMILES of *24340347_Intermediate* (extracted from the docking pose) to identify close analogues in ZINC and related collections. The goal was to assess scaffold prevalence and synthetic accessibility. The search returned a commercially available compound with a Tanimoto similarity of $\approx 93\%$, differing primarily in double-bond stereochemistry.

To determine stereochemistry sensitivity, both the simplified core of the original hit and its stereochemical analogue were re-docked into the S2 pocket using SwissDock for consistency. The docking free energies were nearly indistinguishable: the hit core scored **-5.126 kcal/mol** and the analogue **-5.081 kcal/mol**, within expected scoring variance, indicating that configuration at the double bond had minimal impact on binding. Both retained predicted key interactions with Arg104 and Asp328.

Physicochemical comparison via SwissADME showed the original core possessed an iLOGP of 1.28 versus 1.31 for the analogue, making the core marginally more hydrophilic—a potential advantage for solubility. The core itself was a low-weight fragment (136 Da) lacking aromatic rings, making it an attractive starting scaffold: it already engages key residues (Asp328 and possibly Arg104) while leaving room for growth.

Accordingly, we designated the core as the fragment scaffold for optimization. It complies with fragment-based criteria (low molecular weight, minimal polar complexity) and provides clear attachment vectors for enhancing binding while preserving efficiency.

Figure 6 depicts the structure of *24340347_Intermediate* with the retained core scaffold annotated; the core is a bicyclic di-nitrogen heterocycle (SMILES: `N/N=C1\C=C[C@H]2OCC=C12`) capable of salt-bridging to Asp328 and forming favorable interactions with Arg104.

Fragment Growing with DeepFrag:

Using ProteinDance's fragment elaboration interface, the hit core's docking pose was analyzed to select an optimal growth vector. One terminus was engaged in essential contacts (Asp328 and Arg104) and thus preserved, while the opposite sp^2 carbon projecting toward an unoccupied vestibular subpocket was annotated as the attachment point (marked with a placeholder "*"). The protein–fragment complex was submitted to DeepFrag, requesting suggestions for appendages at this site. DeepFrag returned candidate fragments with predicted contribution scores; the top five suggestions were evaluated by appending them to the core and re-docking, with their docking affinities and ligand efficiencies computed (Table 2).

Table 2 Top fragment additions suggested by DeepFrag for the hit core (attached at the designated site). Docking affinity (SwissDock ΔG) and ligand efficiency (LE) after attachment are reported. LE is defined as binding free energy per heavy atom.

Fragment added	Docking Affinity (kcal/mol)	Heavy Atoms	LE (kcal/mol/HA)
$-CH_3$	-5.21	11	0.47
$-CH_2OH$	-5.26	12	0.44
–CHO	-5.20	12	0.43
$-CH_2CH_3$	-5.23	13	0.40
$-C{\equiv}N$	-5.27	12	0.44

The hydroxymethyl and cyano additions provided the largest affinity gains (≈ -5.26 to -5.27 kcal/mol) while preserving ligand efficiency ≈0.44), despite minimal size increase. Between them, the hydroxymethyl fragment was selected based on medicinal chemistry rationale: it introduces an unambiguous hydrogen-bond donor that can engage the local polar network or ordered water molecules, with lower concern for potential liabilities compared to nitriles. The chosen modification was appended to the core using molecular editing tools, yielding the derivative with SMILES `NNC1(CO)C=CC2OCC=C12` ($C_8H_{12}N_2O_2$), retaining neutrality and proper valence.

The modified compound was re-docked using SwissDock, producing an improved binding free energy of approximately **-5.3 kcal/mol**. This modest gain (0.2 kcal/mol) is significant given the addition of a single heavy atom and preserved ligand efficiency (0.44 kcal/mol/HA), indicating a productive optimization step. Structural analysis confirmed the new hydroxymethyl group forms an additional polar contact—likely a

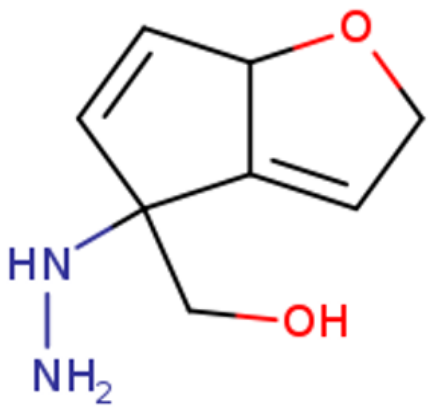

Fig. 7 Optimized lead compound in the S2 pocket, showing the introduced hydroxymethyl interaction (likely with Glu494) alongside preserved contacts to Asp328 and Arg104.

hydrogen bond to Glu494 or a water-mediated bridge—in the S2 pocket, while original anchoring interactions (salt bridge to Asp328 and cation–π proximity to Arg104) remained intact. The optimized molecule was thus designated as the lead compound for further characterization (Figure 7).

3.4 ADMET Prediction and Drug-likeness Evaluation

To contextualize the physicochemical profile of our lead scaffold (SMILES: `NNC1(CO)C=CC2OCC=C12`, $C_8H_{12}N_2O_2$), we compared its key descriptors against those of marketed SSRIs—fluoxetine, paroxetine, sertraline, fluvoxamine—and the endogenous SERT ligand serotonin. As shown in Table 3 and Figure 8, the lead diverges substantially from classic SSRIs in molecular size, polarity and aromatic content:

Table 3 Comparison of key physicochemical properties for the lead compound and marketed SSRIs.

Descriptor	Lead	Fluoxetine	Paroxetine	Sertraline	Fluvoxamine
Molecular weight (Da)	168.2	326.3	223.3	223.8	253.3
LogP	−0.92	5.63	1.36	3.40	0.40
TPSA (Å^2)	67.51	9.23	39.72	3.24	70.00
H-bond acceptors (HBA)	4	1	4	1	4
H-bond donors (HBD)	3	0	1	0	2
Rotatable bonds	2	5	3	3	7
Aromatic rings	0	2	1	1	1

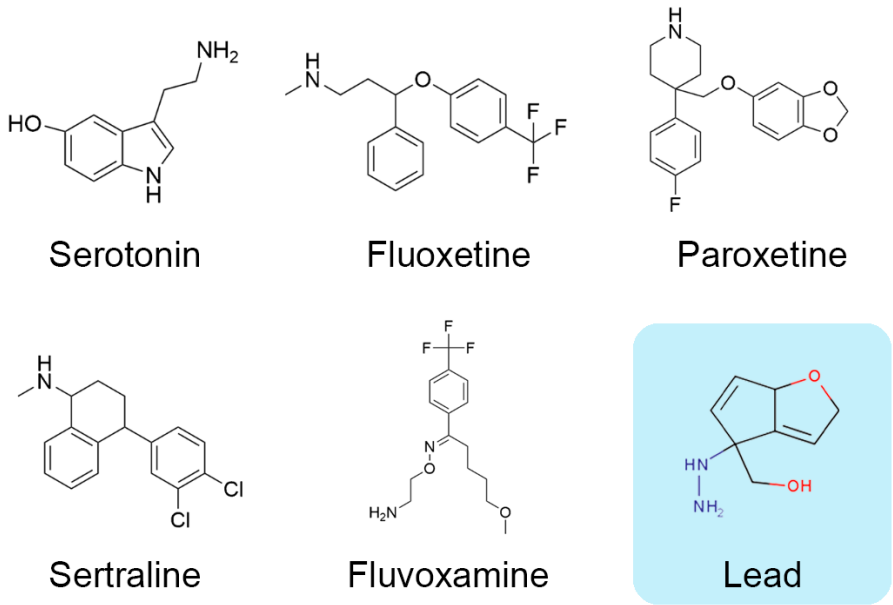

Fig. 8 Comparison with SSRIs Drugs

The lead's low molecular weight (168.2 Da) contrasts with the heavier SSRIs (223–326 Da), a trait that generally enhances aqueous solubility but may compromise binding affinity and half-life. Its highly negative LogP (–0.92) reflects pronounced polarity—beneficial for solubility and GI absorption but suboptimal for passive CNS permeation, which typically requires LogP values between 1–3. The TPSA of 67.5 $Å^2$ is comparable to serotonin (64.9 $Å^2$) yet notably higher than most SSRIs (3.2–39.7 $Å^2$), further suggesting limited blood–brain barrier (BBB) transit. With HBA/HBD counts of 4/3, the scaffold is more polar than fluoxetine and sertraline, aligning with the BOILED-Egg model's prediction of poor BBB penetration and the need for active transport or prodrug approaches.

Moreover, the absence of aromatic rings and low rotatable bond count (2) confer synthetic tractability and conformational rigidity but diverge from the lipophilic, π -rich frameworks observed in SSRIs—features often critical for engaging hydrophobic pockets within SERT. By contrast, marketed compounds feature 1–2 aromatic moieties and 3–7 rotatable bonds, balancing flexibility and target complementarity.

Taken together, our lead exhibits an exemplary solubility and compliance with Lipinski's rules (no violations), yet its excessive polarity and lack of aromaticity underscore a key lead-optimization challenge: introducing nonpolar, aromatic substituents to elevate lipophilicity into the 1–3 range, reduce TPSA below 70 $Å^2$, and enhance passive CNS uptake. These strategic modifications will aim to preserve oral bioavailability while unlocking central SERT engagement, paving the way for downstream *in vitro* and *in vivo* validation.

4 Discussion

4.1 Accelerating Drug Discovery with AI4Science

Pharmaceutical R&D has historically been defined by protracted timelines (often over 12 years from laboratory to approval) and enormous costs (frequently exceeding $ 1–2 billion per new drug), yet with failure rates above 90%. [1] This inefficiency not only squanders resources but also stalls innovation and the promise of precision medicine—where therapies are tailored to specific patient groups. The emerging paradigm of AI for Science (AI4S) seeks to reverse this trend by embedding advanced machine-learning models into each stage of the scientific workflow. In the present study, we exemplified this approach through ProteinDance, an integrated AI-driven pipeline that automates target identification, structure modeling, binding-pocket prediction, de novo molecule generation, and in silico toxicity filtering. By uniting state-of-the-art tools (e.g. AlphaFold for structure, P2Rank for pocket detection, transformers for molecule design) under an AI-guided decision framework, ProteinDance dramatically compresses the early drug discovery process. The successful identification of a novel serotonin transporter modulator in silico – from target selection to lead molecule – underscores how such AI4S pipelines can reduce upfront experimentation and accelerate the path from "disease to drug." Notably, recent AI-driven projects have reported unprecedented speed in discovery. For example, a deep generative model discovered potent kinase inhibitors in only 21 days [22], and during the COVID-19 pandemic an AI system narrowed 2,000 candidate antibodies down to a lead that entered clinical trials in just three months. These cases illustrate that AI-enabled workflows can now accomplish in weeks what traditionally took years, fundamentally redefining the pace of early drug development.

4.2 Toward Low-Cost Precision Medicine

The convergence of AI and biomedical science is likewise paving the way for more cost-efficient, patient-tailored therapies. By automating design and decision-making, AI-driven pipelines slash the costs of exploring chemical and genomic space, making it feasible to pursue treatments for smaller patient populations and rare diseases that were previously neglected due to economic constraints[24]. For instance, only about 5–7% of the 7,000 known rare diseases currently have an FDA-approved therapy – a consequence of traditional R&D being too slow and expensive to address ultra-niche conditions. Lowering discovery costs with AI changes this calculus. As our ProteinDance demonstration suggests, a focused in silico campaign can yield a viable lead for a specific molecular target (here, an allosteric site on SERT) with minimal wet-lab work, empowering researchers to tackle more specialized mechanisms or personalized interventions. In the context of depression treatment, this means one could rapidly prototype drugs for novel targets (or novel binding sites on old targets) that promise fewer side effects or faster onset than existing SSRIs – a level of iterative exploration that would have been prohibitively costly before. More broadly, AI systems are beginning to integrate vast patient datasets (genomic, proteomic, clinical) to identify patient subgroups and predict differential drug responses. This synergy between AI-driven drug

design and AI-driven patient stratification forms a backbone of precision medicine: not only can compounds be designed with specific molecular properties, but they can be matched to the patients most likely to benefit, yielding more effective treatments with fewer trial-and-error failures. Indeed, early evidence shows AI can expedite the development of more precise treatments with fewer side effects at far lower cost than the traditional one-size-fits-all approach [24]. Our results contribute to this vision by showing that a modular AI pipeline can deliver a targeted lead compound (for a well-characterized psychiatric protein) in silico, ready for refinement and personalization in subsequent studies.

4.3 Limitations

Several constraints temper the current study's conclusions. First, the pipeline's predictions remain unvalidated in vitro or in vivo; the absence of wet-lab assays leaves binding affinity, functional activity, and off-target effects unconfirmed, limiting immediate translational confidence. Second, while we propose re-optimizing historical "failed" compounds, our present work does not empirically demonstrate this capability at scale; future large-batch experiments are required to quantify success rates, cost savings, and safety improvements. Third, although the transformer backbone delivers competitive performance, it has not yet achieved an optimal cost-to-quality equilibrium—model size, inference latency, and fine-tuning expense still pose barriers that warrant algorithmic compression and data-efficient training strategies. Finally, ProteinDance operates through discrete tool invocations orchestrated by a single controller; building a fully autonomous multi-agent chain that can reason, plan, and adaptively call domain-specific services would enhance robustness, reduce human oversight, and broaden applicability to more complex discovery tasks.

4.4 Future Outlook and Applications

Looking ahead, the integration of AI into drug discovery is poised to not only accelerate new projects but also resurrect and optimize prior drug candidates that would otherwise be lost. Despite the time and money invested, roughly 90 % of drug candidates fail in clinical trials and are shelved by developers. [25] Traditionally, it has been "easier to simply throw away a failing drug than to fix it" due to the enormous experimental effort required to address issues like off-target toxicity. However, modern AI offers a powerful antidote to this attrition. Drug repositioning and rescue strategies leverage AI to analyze existing compounds (including those that failed or were abandoned) and identify new opportunities for them. [26] By rapidly sifting through vast chemical and biomedical data, AI can uncover hidden drug–target connections or design analogues that remedy the original compound's flaws. In fact, several startups are now actively acquiring "failed" drug assets to apply AI-driven redesign—fixing their toxicity or efficacy issues in silico—and then re-entering them into trials. This approach is not merely theoretical: it is emerging as a prudent, cost-effective extension of drug development, effectively giving shelved molecules a second chance at success. We envisage that pipelines like ProteinDance could be deployed in such batch re-optimization efforts. For example, one could take a set of published drug

candidates that showed therapeutic promise but were dropped for toxicity, and run each through the ProteinDance workflow (or a similar transformer-based generative model) to suggest structural modifications that reduce toxicity while retaining efficacy. Early in silico toxicity filters and property optimizations would winnow the set of analogues, after which only the most promising redesigned molecules would undergo focused experimental testing. By iterating this cycle, it may be possible to salvage prior R& D investments at a fraction of the cost of de novo development. Importantly, unlike traditional pipelines that often ignore negative data, AI models can actively learn from failed project data to inform future designs. [27] The lessons encoded in past failures – for instance, knowing which substructures caused off-target effects or poor pharmacokinetics – become fuel for algorithms that propose better candidates in the future. Harnessing this "dark matter" of pharmaceutical research (the vast archive of compounds that didn't make it) could substantially raise the productivity of drug discovery moving forward.

Beyond rescuing old leads, there are myriad other future directions for AI-augmented drug design. One promising area is the creation of closed-loop discovery systems: coupling generative design platforms like ProteinDance with automated synthesis and high-throughput biological testing. Such an integration would enable rapid experimental feedback to retrain models, creating a self-refining cycle of hypothesis and validation. Early steps in this direction have already been taken – for example, AI "virtual labs" have iteratively optimized biologics and mRNA vaccine constructs in months – and extending similar principles to small-molecule discovery is a logical next step. Another avenue is expanding the scope of AI4Science pipelines to other therapeutic modalities (e.g. peptide drugs, gene therapies) and to multi-target polypharmacology. [28] [29] [30] In each case, the core concept is the same: incorporate knowledge-driven AI at every decision point to guide experiments more efficiently than human intuition alone. Challenges remain, of course – from ensuring data quality and model interpretability to navigating regulatory acceptance of AI-designed molecules – but the trajectory is clear. By maximizing "knowledge reuse" and minimizing trial-and-error, AI stands to transform pharmaceutical research from a high-risk gamble into a more deterministic, low-cost innovation engine.

In summary, our work demonstrates that an AI-integrated, structure-based pipeline can produce a credible drug lead in silico, exemplifying the broader trend of AI-driven precision medicine. The ProteinDance framework merges physics-based modeling with generative AI and domain expertise, yielding a streamlined path from a disease hypothesis to a candidate molecule. Such approaches directly address long-standing bottlenecks in drug development, from the identification of novel targets and allosteric sites to the optimization of lead compounds with improved safety profiles. We anticipate that continued refinement of these technologies – and their application both to new diseases and to "lost" compounds from previous decades – will markedly increase the productivity of drug discovery. Ultimately, AI-driven pipelines offer a template for future research in which life-saving therapies can be developed faster, cheaper, and with greater precision than ever before, inaugurating a new era of truly personalized and efficient medicine.

References

[1] Sertkaya, A., Beleche, T., Jessup, A. & Sommers, B. D. Costs of drug development and research and development intensity in the us, 2000-2018. *JAMA Network Open* **7** (2024). URL https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2820562.

[2] Van Noorden, R. & Perkel, J. M. Ai and science: what 1,600 researchers think. *Nature* **621**, 672–675 (2023). URL https://www.nature.com/articles/d41586-023-02980-0.

[3] Abramson, J. *et al.* Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature* **630**, 493–500 (2024). URL https://www.nature.com/articles/s41586-024-07487-w.

[4] El Arab, R. A. *et al.* Artificial intelligence in vaccine research and development: an umbrella review. *Frontiers in Immunology* **16** (2025).

[5] Zhang, H. *et al.* Algorithm for optimized mrna design improves stability and immunogenicity. *Nature* **621**, 1–3 (2023). URL https://www.nature.com/articles/s41586-023-06127-z.

[6] Zhang, R., Wen, H., Lin, Z., Li, B. & Zhou, X. Artificial intelligence-driven drug toxicity prediction: Advances, challenges, and future directions. *Toxics* **13**, 525 (2025). URL https://www.mdpi.com/2305-6304/13/7/525.

[7] Ding, L., Wu, Z., Wu, Q., Wei, R. & Li, E. Prevalence and lifestyle determinants of depressive symptoms among chinese children and adolescents. *Scientific Reports* **14** (2024). URL https://www.nature.com/articles/s41598-024-78436-w.pdf.

[8] Bartlett, E. A. *et al.* Serotonin transporter binding in major depressive disorder: impact of serotonin system anatomy. *Molecular Psychiatry* **27** (2022).

[9] Nagayasu, K. Serotonin transporter: Recent progress of in silico ligand prediction methods and structural biology towards structure-guided in silico design of therapeutic agents. *Journal of Pharmacological Sciences* **148**, 295–299 (2022). URL https://www.sciencedirect.com/science/article/pii/S1347861322000044.

[10] Consortium, T. U. Uniprot: a hub for protein information. *Nucleic Acids Research* **43**, D204–D212 (2014).

[11] Burley, S. K. *et al.* Rcsb protein data bank: biological macromolecular structures enabling research and education in fundamental biology, biomedicine, biotechnology and energy. *Nucleic Acids Research* **47**, D464–D474 (2018).

[12] Krivák, R. & Hoksza, D. P2rank: machine learning based tool for rapid and accurate prediction of ligand binding sites from protein structure. *Journal of Cheminformatics* **10** (2018).

[13] Gaulton, A. *et al.* Chembl: a large-scale bioactivity database for drug discovery. *Nucleic Acids Research* **40**, D1100–D1107 (2011).

[14] Radford, A. *et al.* Language models are unsupervised multitask learners. *OpenAI blog* **1**, 9 (2019).

[15] Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S. & Hopkins, A. L. Quantifying the chemical beauty of drugs. *Nature chemistry* **4**, 90–98 (2012). URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3524573/.

[16] Li, Y. Deep reinforcement learning: An overview. *arXiv (Cornell University)* (2017).

[17] Olivecrona, M., Blaschke, T., Engkvist, O. & Chen, H. Molecular de-novo design through deep reinforcement learning. *Journal of Cheminformatics* **9** (2017).

[18] Landrum, G. *et al.* rdkit/rdkit: 2025_03_5 (q1 2025) release. *Zenodo* (2025). URL https://zenodo.org/records/16439048.

[19] Coleman, J. A., Green, E. M. & Gouaux, E. X-ray structures and mechanism of the human serotonin transporter. *Nature* **532**, 334–339 (2016). URL https://www.nature.com/articles/nature17629.

[20] Plenge, P. *et al.* Steric hindrance mutagenesis in the conserved extracellular vestibule impedes allosteric binding of antidepressants to the serotonin transporter. *The Journal of Biological Chemistry* **287**, 39316–39326 (2012). URL https://pubmed.ncbi.nlm.nih.gov/23007398/.

[21] Andersen, J. *et al.* Molecular basis for selective serotonin reuptake inhibition by the antidepressant agent fluoxetine (prozac). *Molecular Pharmacology* **85**, 703–714 (2014).

[22] Zhavoronkov, A. *et al.* Deep learning enables rapid identification of potent ddr1 kinase inhibitors. *Nature Biotechnology* **37**, 1038–1040 (2019). URL https://www.gwern.net/docs/rl/2019-zhavoronkov.pdf.

[23] Grosdidier, A., Zoete, V. & Michielin, O. Swissdock, a protein-small molecule docking web service based on eadock dss. *Nucleic Acids Research* **39**, W270–W277 (2011).

[24] Huang, K. *et al.* A foundation model for clinician-centered drug repurposing. *Nature Medicine* **30**, 1–13 (2024). URL https://www.nature.com/articles/s41591-024-03233-x.

[25] Sun, D., Gao, W., Hu, H. & Zhou, S. Why 90% of clinical drug development fails and how to improve it? *Acta Pharmaceutica Sinica B* **12** (2022).

[26] Yin, Z. & Wong, S. T. C. Artificial intelligence unifies knowledge and actions in drug repositioning. *Emerging Topics in Life Sciences* **5**, 803–813 (2021).

[27] Masarone, S. *et al.* Advancing predictive toxicology: Overcoming hurdles and shaping the future. *Digital Discovery* **4** (2024).

[28] Cheng, J. *et al.* Accurate proteome-wide missense variant effect prediction with alphamissense. *Science* **381** (2023). URL https://www.science.org/doi/10.1126/science.adg7492.

[29] Rehman, A. U. *et al.* Role of artificial intelligence in revolutionizing drug discovery. *Fundamental Research* **5** (2024). URL https://www.sciencedirect.com/science/article/pii/S266732582400205X.

[30] Vora, L. K. *et al.* Artificial intelligence in pharmaceutical technology and drug delivery design. *Pharmaceutics* **15**, 1916–1916 (2023). URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10385763/.